Worksheet: AI Face Tracking Taster

What you'll make today

1. **Part 1 — "Security Console"**: webcam + AI face landmarks + mugshot strip + save images (cut and paste code).
2. **Part 2 — "Texture Mask"**: pick a face mask texture (you edit PNG images in Paint, no coding!).

What you need

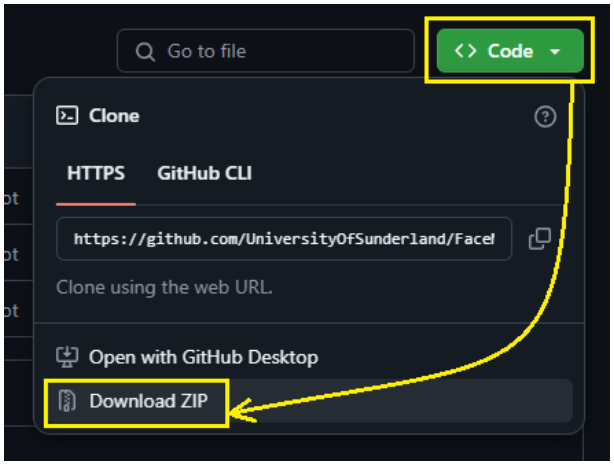
- Chrome (recommended)
- VS Code
- Webcam
- The provided zip project folder

Stop rule: If something doesn't match the screenshot after 60 seconds, put your hand up.

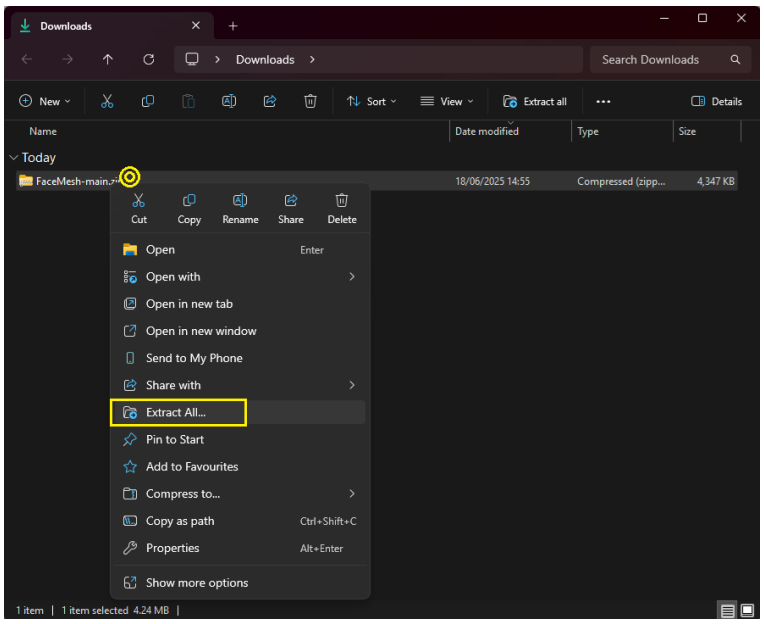
Part 0: Setup (Do this once)

Step 0.1 Download + Extract

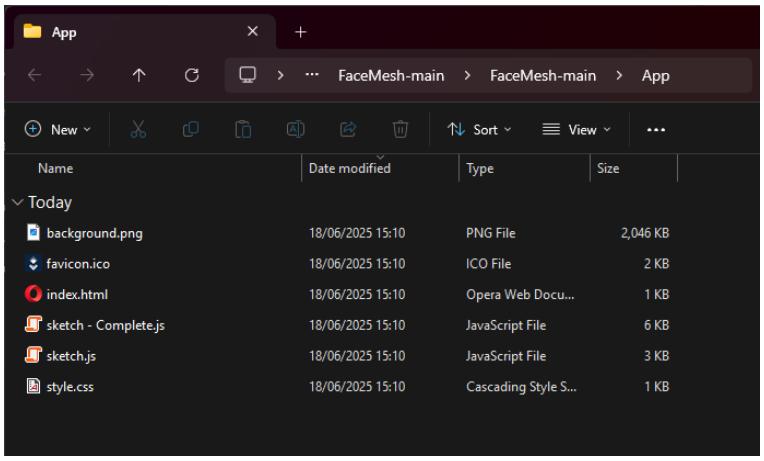
1. Download the activity zip from GitHub (link below).
   https://github.com/UniversityOfSunderland/FaceMesh



2. Locate the downloaded zip, right-click → Extract All → Extract.

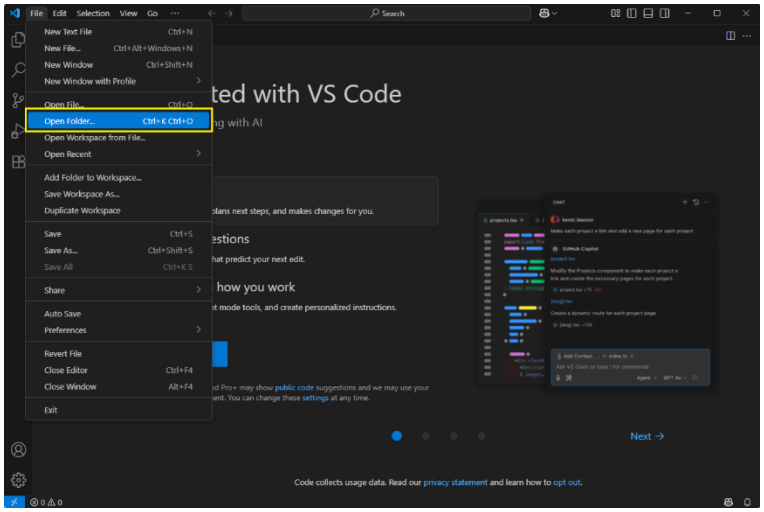


3. Open the extracted folder and confirm you can see files like:
    - index.html
    - sketch.js
    - style.css

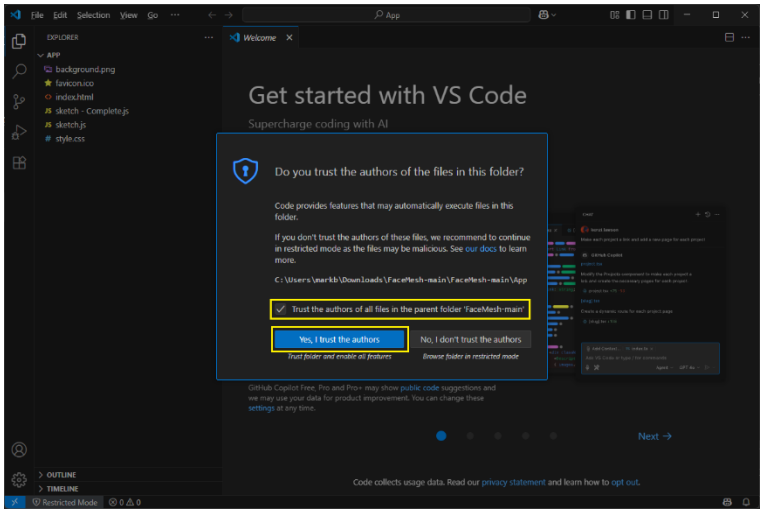


Step 0.2 Open in VS Code (avoid "wrong folder" bug)

- Open VS Code
- **File → Open Folder...**
- Select the extracted **App** folder (or the folder that contains index.html)

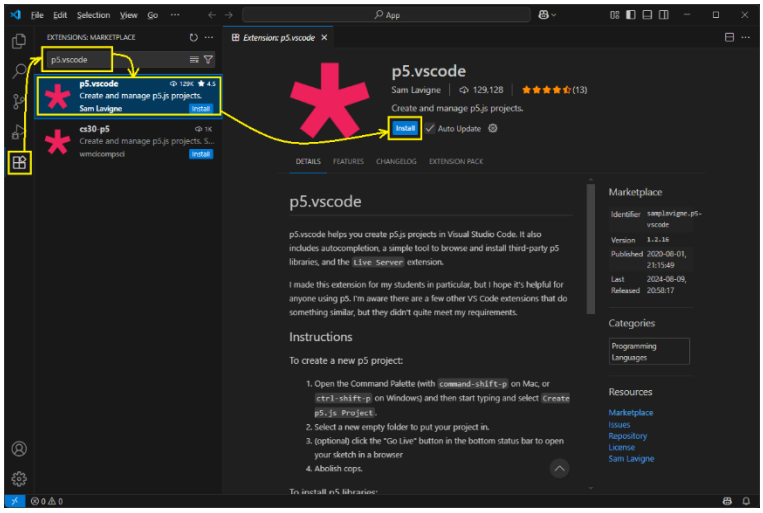


- Then Select “Yes, I trust the authors”.



Step 0.3 Install ‘p5.vscode’

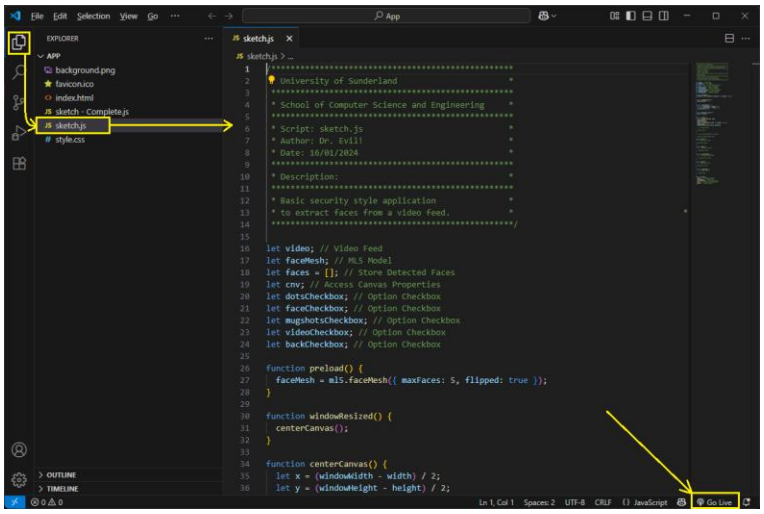
- Navigate to the Extensions icon in the side menu.
- Search for ***“p5.vscode”***, select and click the blue **Install** button.



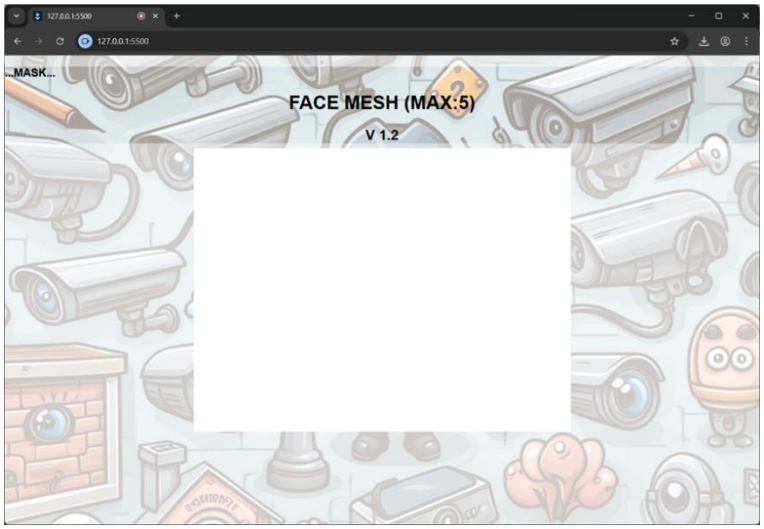
- Select to **“Trust Publishers & Install”** if prompted.

Step 0.4 Run the website

- Bottom-right in VS Code: click **Go Live**



- You may need to **Allow** permissions on the first time running this.
- Chrome opens with a page that shows the canvas area.
- **"Allow while visiting the site"** when requested for use of the camera.



Part 1: Build the "Security Console"

Goal

You will add features in a safe order:

1. Menu UI
2. Video feed
3. Background toggle
4. Face mesh dots
5. Face isolation
6. Mugshots strip
7. Save images (press keys)

Golden rule

After each step:

- **CTRL+S**, then look at the browser.
- If nothing changes, refresh once.
- If still nothing, ask for help.

1.1: Menu Checkboxes

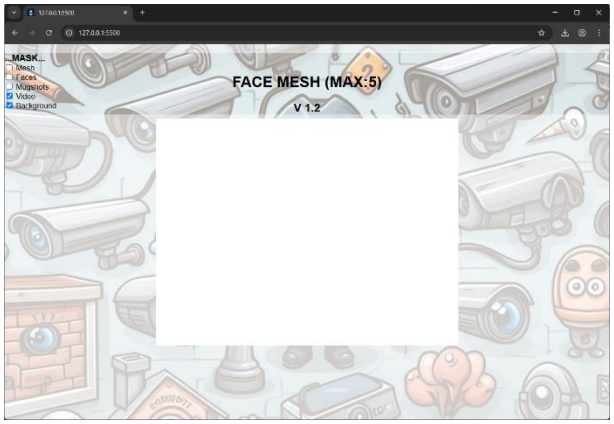
Open **sketch.js** and find:

```
// 1 - Menu (Start)
// PASTE STEP 1 CODE HERE
// 1 - Menu (End)
```

Paste the menu code (see Code below).

```
// 1 - Menu (Start)
dotsCheckbox = createCheckbox(" Mesh");
dotsCheckbox.position(0, 40);

faceCheckbox = createCheckbox(" Faces");
faceCheckbox.position(0, 60);

mugshotsCheckbox = createCheckbox(" Mugshots");
mugshotsCheckbox.position(0, 80);

videoCheckbox = createCheckbox(" Video", true);
videoCheckbox.position(0, 100);

backCheckbox = createCheckbox(" Background", true);
backCheckbox.position(0, 120);
// 1 - Menu (End)
```

Checkpoint: You can tick “Video / Background / Mesh / Faces / Mugshots”.



1.2: Video Feed

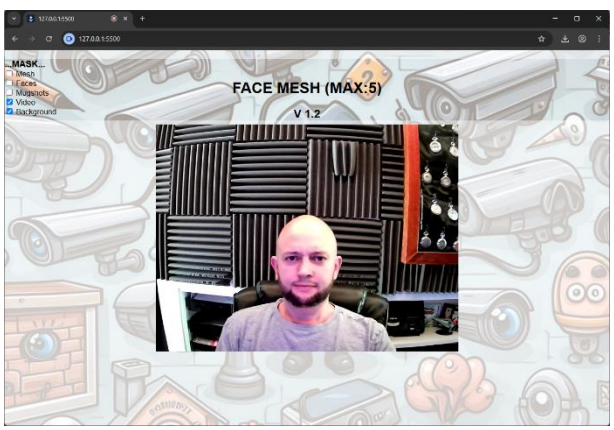
Find:

```
// Step 2
function drawVideo() {
  // 2 - Video (Start)
  // PASTE STEP 2 CODE HERE
  // 2 - Video (End)
}
```

Add the video draw code.

```
  // 2 - Video (Start)
  if (videoCheckbox.checked()) {
    image(video, 0, 0);
  }
  // 2 - Video (End)
```

Checkpoint: You can see yourself when “Video” is ticked.



1.3: Background toggle

Find:

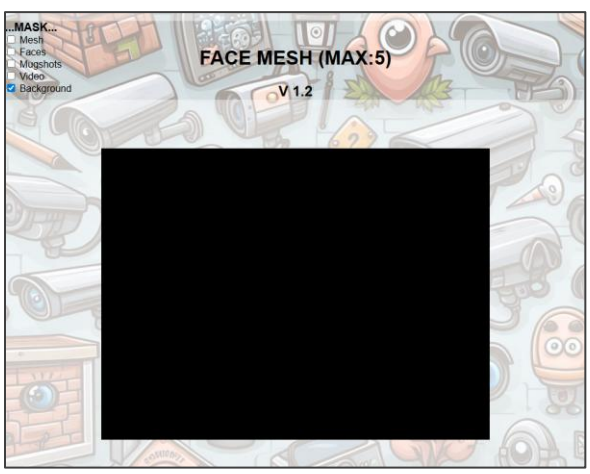
```
// Step 3
function drawBackground() {
  // 3 - Background (Start)
  // PASTE STEP 3 CODE HERE
  // 3 - Background (End)
}
```

Add the background code.

```
  // 3 - Background (Start)
  if (backCheckbox.checked()) {
    background(0);
  }
  // 3 - Background (End)
```

Checkpoint:

- Background ON + Video OFF = black canvas
- Background OFF + Video OFF = you may see still frames (that's fine)



1.4: Mesh dots

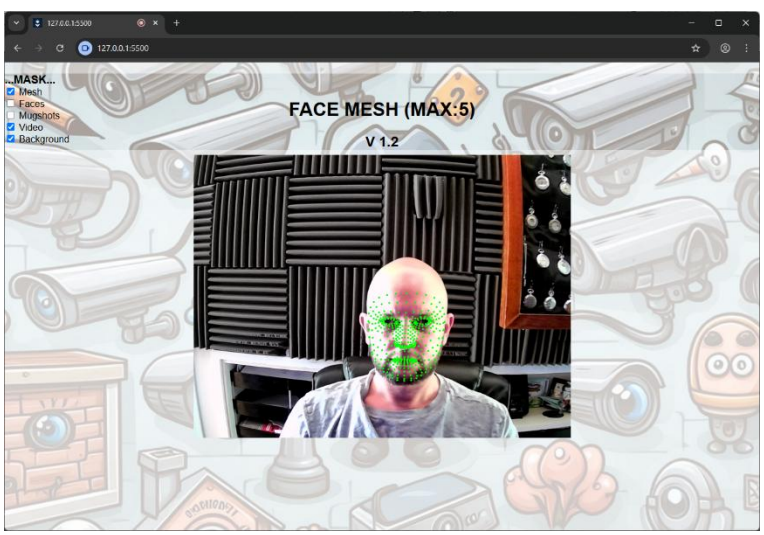
Find:

```
// Step 4
function drawMeshDots() {
  // 4 - Mesh Dots (Start)
  // PASTE STEP 4 CODE HERE
  // 4 - Mesh Dots (End)
}
```

Add the mesh dots code.

```
  // 4 - Mesh Dots (Start)
  if (dotsCheckbox.checked() && faces.length > 0) {
    for (let f = 0; f < faces.length; f++) {
      const face = faces[f];
      noStroke();
      fill(0, 255, 0);
      for (let p = 0; p < face.keypoints.length; p++) {
        circle(face.keypoints[p].x, face.keypoints[p].y, 3);
      }
    }
  }
  // 4 - Mesh Dots (End)
```

Checkpoint: Green points appear on your face when “Mesh” is ticked.



1.5: Isolate faces

Find:

```
// Step 5
function isolateFaces() {
  // 5 - Isolate Faces (Start)
  // PASTE STEP 5 CODE HERE
  // 5 - Isolate Faces (End)
}
```

Add isolate code.

```
  // 5 - Isolate Faces (Start)
  if (faceCheckbox.checked() && faces.length > 0) {
    for (let f = 0; f < faces.length; f++) {
      const frame = video.get(0, 0, width, height);
      const face = faces[f];
      const box = face.box;

      const shape = createGraphics(width, height);
      shape.fill(0);
      shape.strokeWeight(5);
      shape.beginShape();

      for (let p = 0; p < face.faceOval.keypoints.length; p++) {
        const k = face.faceOval.keypoints[p];
        shape.point(k.x, k.y);
        shape.curveVertex(k.x, k.y);
      }

      shape.endShape(CLOSE);
      frame.mask(shape);

      const faceImage = frame.get(box.xMin, box.yMin, box.width, box.height);
      image(faceImage, box.xMin, box.yMin);
    }
  }
  // 5 - Isolate Faces (End)
```

Checkpoint: Turning on “Faces” produces a cut-out face region. Play around with Video and Background On/Off.

1.6: Mugshots strip

Find:

```
// Step 6
function drawMugshots() {
  // 6 - Mugshots (Start)
  // PASTE STEP 6 CODE HERE
  // 6 - Mugshots (End)
}
```

Add the mugshot code.

```
  // 6 - Mugshots (Start)
  if (mugshotsCheckbox.checked()) {
    const bannerHeight = 145;

    strokeWeight(6);
    stroke(125);
    fill(255);
    rect(3, 3, width - 6, bannerHeight, 5);

    for (let f = 0; f < faces.length; f++) {
      const face = faces[f];
      const box = face.box;

      const faceImage = video.get(box.xMin, box.yMin, box.width, box.height);
      faceImage.resize(80, 100);

      image(faceImage, (f * 85) + 10, 10);
      noStroke();
      fill(0);
      textSize(18);
      text("face " + f, (f * 85) + 25, 130);
    }
  }
  // 6 - Mugshots (End)
```

Checkpoint: Small face thumbnails appear across the top.

1.7: Save images (keyboard)

Find:

```
// Step 7 - Saving
// 7 - Save (Start)
// PASTE STEP 7 CODE HERE
// 7 - Save (End)
```

Add the save blocks.

```
// 7 - Save (Start)
function saveFaces() {
  if (mugshotsCheckbox.checked()) {
    for (let f = 0; f < faces.length; f++) {
      const box = faces[f].box;
      const faceImage = video.get(box.xMin, box.yMin, box.width, box.yMin ?
box.height : box.height);
      // (keep it simple; box.height exists)
      faceImage.save("face" + f, "png");
    }
  }
}

function saveSnapshot() {
  saveCanvas("canvas.png");
}

function keyPressed() {
  if (key === "m" || key === "M") saveFaces();
  if (key === "s" || key === "S") saveSnapshot();
}
// 7 - Save (End)
```

- Press **M** to save faces (mugshots must be enabled)
- Press **S** to save the whole canvas

Checkpoint: Your browser downloads PNG files.

Part 1 Common Fixes

Nothing shows in browser

- Is Go Live running? (bottom bar says “Port: ####”)
- Did you open the correct folder (the one with index.html)?

Camera doesn’t work

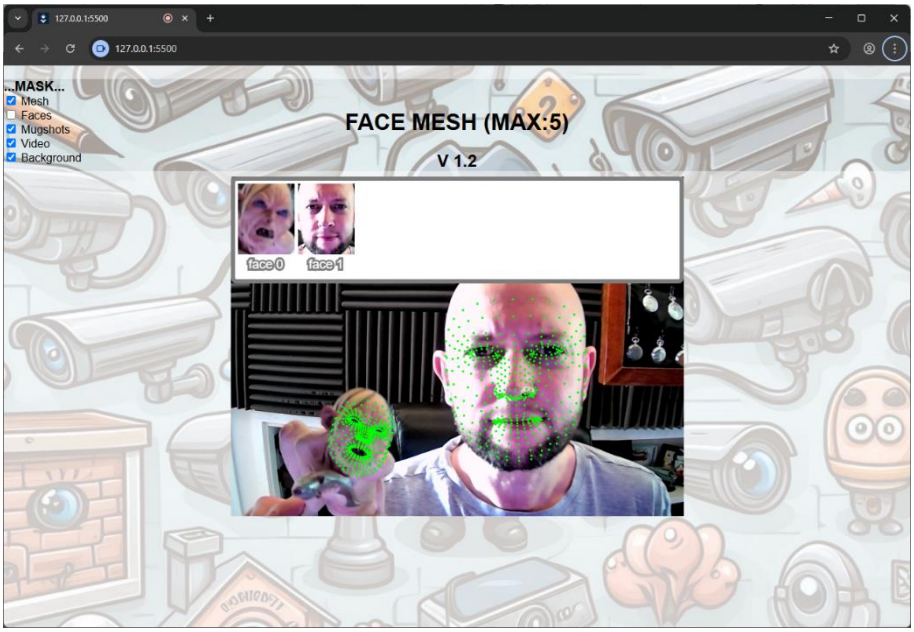
- Chrome address bar → camera icon → Allow
- Refresh page

No green dots

- Tick **Mesh**
- Get your face in frame, good light

It worked then stopped

- Deleted a brace **}** - compare with ***sketch - Complete.js***.



Part 2 Steps

2.1 Download, Unzip, Open and Test 'Portraits'.

- Download the activity zip from GitHub (link below).
  https://github.com/UniversityOfSunderland/Portraits
- Unzip, **Open Folder** in VS Code and **Go Live** as with Part 1.

Checkpoint: Tick:

- Video
- Dots
- Mask
  and select Mask 0/1/2/3/All.

Step 2.2 Edit a Mask in Paint

1. In the folder, right-click mask0.png → **Open with** → **Paint**
2. Draw something bold (high contrast works best)
3. Save (CTRL+S)

Checkpoint: Refresh browser → Mask updates.

Step 2.3 Quick "Design Recipe"

In Paint:

- Use **big shapes**, not thin lines
- Use **high contrast**
- Keep important detail in the **center** of the image (it maps onto the face)
- Consider using photos and images from the Internet.
  - Make a mask of a famous person.
  - Make a superhero mask.
  - Make a troll mask.

Step 2.4 Save a Screenshot of Results

In this app:

- Press **S** to save the canvas (Portraits.png)

Checkpoint: A file downloads.